

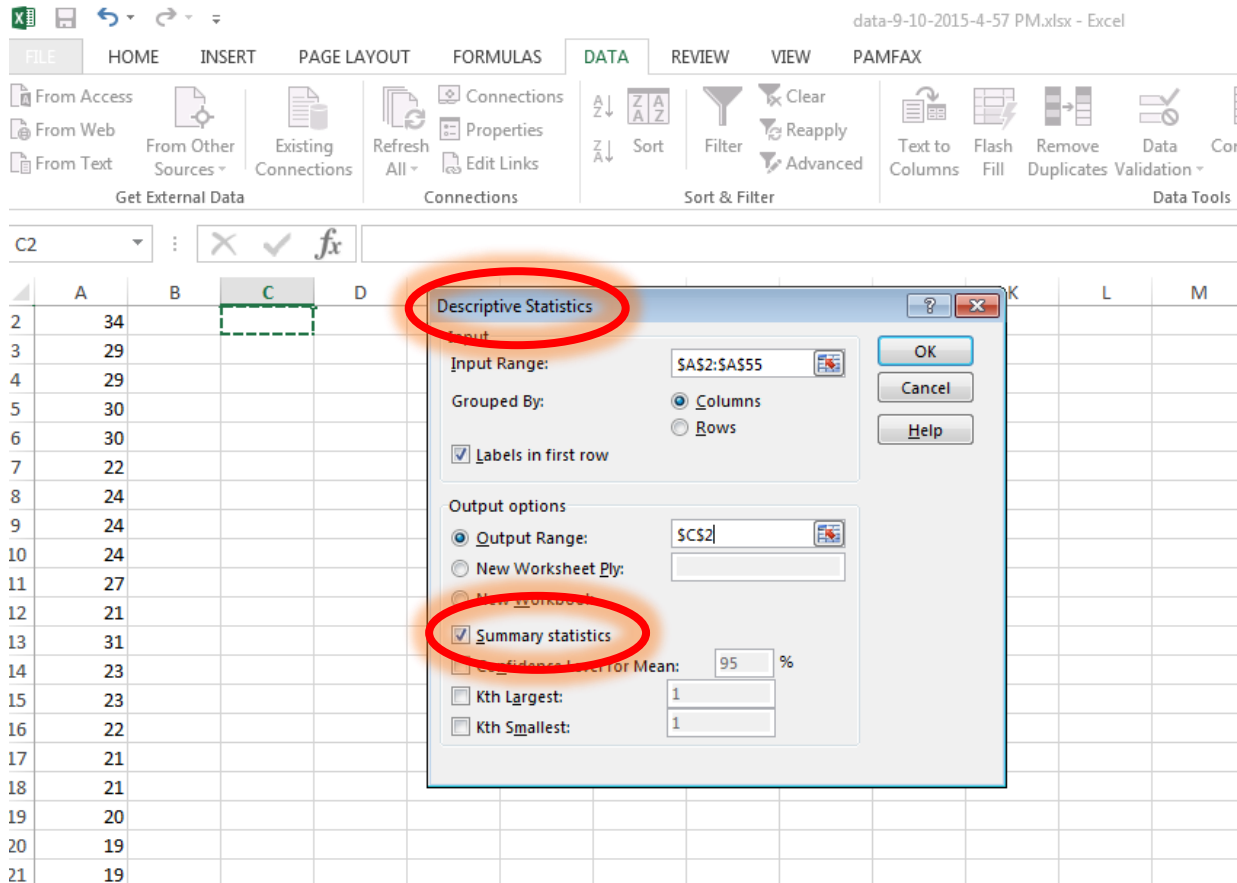
Descriptive Statistics

Excel has functions for almost all the descriptive statistics that we learned. Suppose our data are located in column A, rows 2 through 55. Then you can get descriptive statistics by the following functions --



Statistic	Excel function
Mean	=AVERAGE(A2:A55)
Median	=MEDIAN(A2:A55)
Mode	=MODE(A2:A55)
Variance	=VAR(A2:A55)
Standard deviation	=STDEV(A2:A55)
1 st quartile Q1	=QUARTILE(A2:A55,1)
3 rd quartile Q3	=QUARTILE(A2:A55,3)
The smallest value	=MIN(A2:A55)
The largest value	=MAX(A2:A55)
Interquartile range IQR	=QUARTILE(A2:A55,3)- QUARTILE(A2:A55,1)
Range	=MAX(A2:A55)- MIN(A2:A55)
Sample size (n)	=COUNT(A2:A55)

Most of these statistics can also be obtained in 1 click, from the Data Analysis Toolpak, where you choose “Descriptive Statistics” and check the box “Summary Statistics”.



Z-scores

To obtain the entire column of Z-scores, let's write a formula

$$Z = \frac{X - \bar{X}}{S}$$

for each observation X . We'll write it only once and then drag over the whole data set. But when we drag through the rows, we want to preserve the sample mean and sample variance from being dragged, so we write \$ in front of their row locations, as below.

data-9-10-2015-4-57 PM.xlsx - Excel

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	A	B	C	D	E	F	G	H	I
1	Data	Z-scores							
2	34	$=(A2-G\$3)/G\7							
3	29					Mean	13.8418		
4	29					Standard Error	1.30196		
5	30					Median	13.9231		
6	30					Mode	24		
7	22					Standard Deviation	9.4784		
8	24					Sample Variance	89.8401		
9	24					Kurtosis	-1.07018		
10	24					Skewness	0.04966		
11	27					Range	33.0769		

Then drag the formula down till the end of the data set. To round to the desired number of decimal digits, use the 00->0 button under the HOME tab

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	A	B	C	D	E	F	G	H	I
1	Data	Z-scores							
2	34	2.126751							
3	29	1.599236				Mean	13.8418		
4	29	1.599236				Standard Error	1.30196		
5	30	1.704739				Median	13.9231		
6	30	1.704739				Mode	24		
7	22	0.860715				Standard Deviation	9.4784		

or format the highlighted cells to a “Number” specifying the desired number of decimal digits. Highlight the column, right-click, choose “Format”, and in the box, select “Number” and “decimal places”.

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Clipboard Font Alignment Number Styles

B2 X Y fx $\text{=(A2-G\$3)/G\$7}$

	A	B	C	D	E	F
1	Data	Z-scores				
2	34	2.126751				
3	29	1.599236				
4	29	1.599236				
5	30	1.704739				
6	30	1.704739				
7	22	0.860715				
8	24	1.071721				
9	24	1.071721				
10	24	1.071721				
11	27	1.38823				

Format Cells

Number Alignment Font Border Fill Protection

Category: General Number Currency Accounting Date Time Percentage Fraction Scientific Text Special Custom

Sample 2.12

Decimal places: 2

Use 1000 Separator (,)

Negative numbers: -1234.10 1234.10 (1234.10) (1234.10)

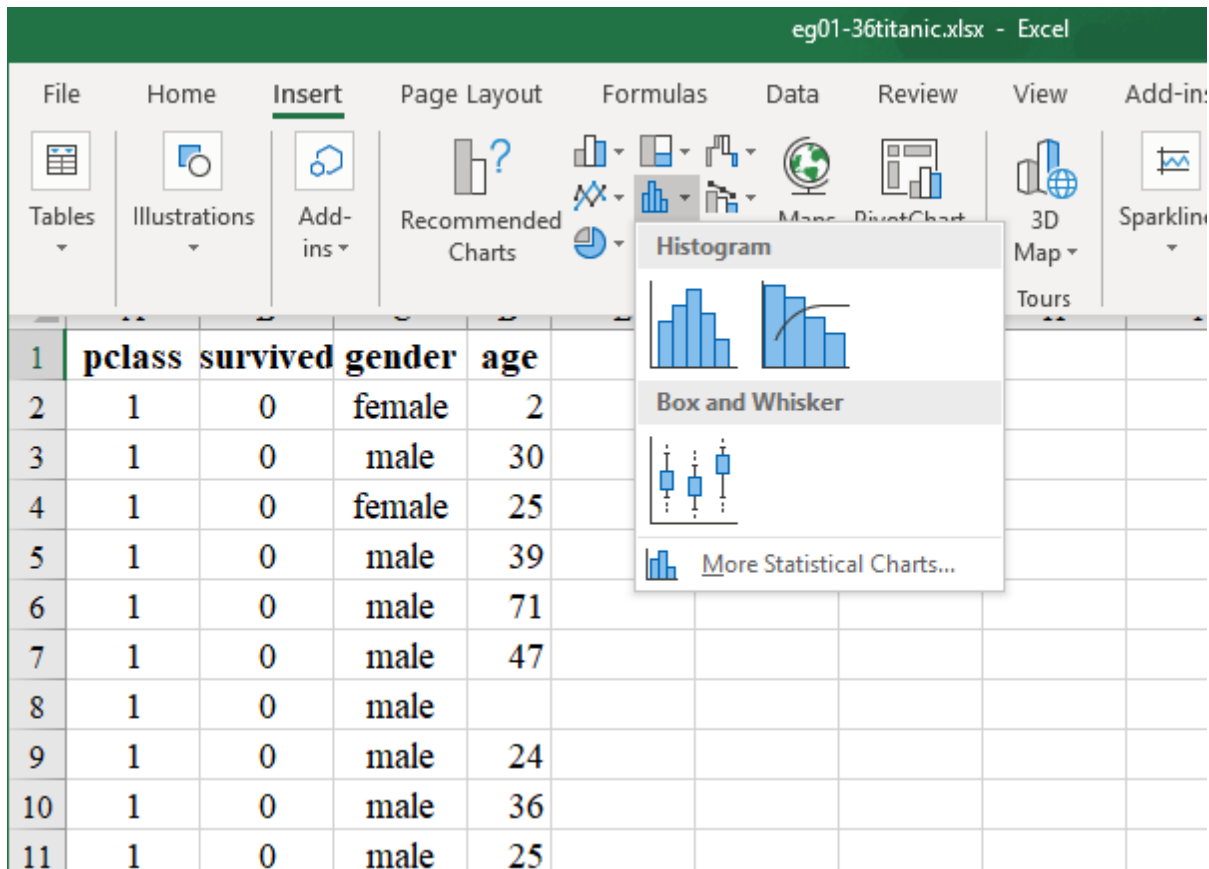
Number is used for general display of numbers. Currency and Accounting offer specialized formatting for monetary value.

OK Cancel

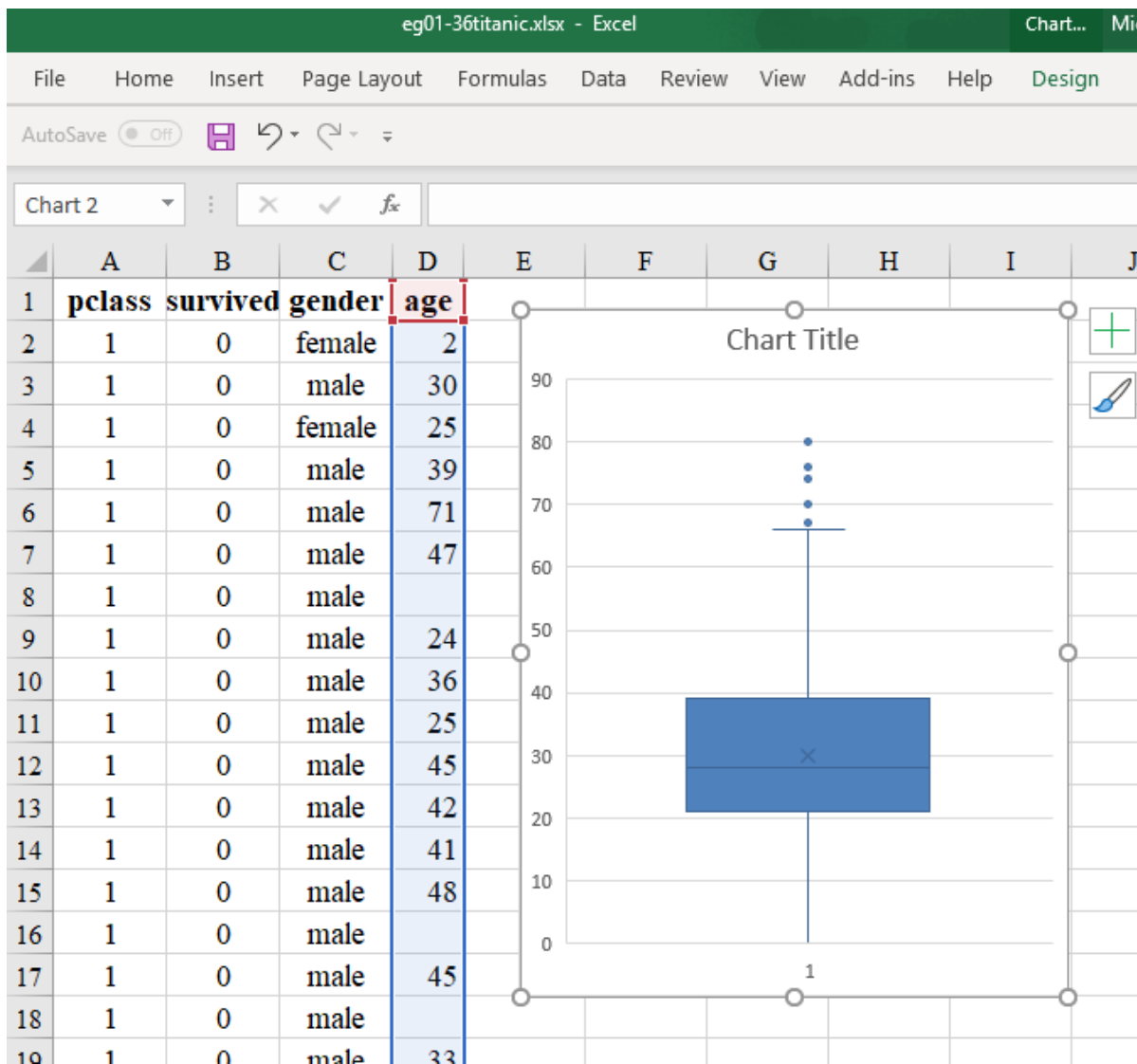
Boxplot

The newest versions of Excel include *boxplots* within the general set of statistical plots.

The drop-down list of *Statistics Charts* is under the Insert tab, next to scatterplots, histogram, and pie charts. In this list, *Box and Whisker* produces a boxplot.



First, highlight one or several data columns for boxplots, then click Insert, Statistics Charts, Box and Whisker.



Covariance and correlation

Excel has functions `=COVARIANCE.S(A2:A55,B2:B55)` for the sample covariance and `=CORREL(A2:A55,B2:B55)` for the correlation coefficient.

Notice that the Data Analysis Toolpak computes *population* covariance, with n in the denominator instead of $(n-1)$. This corresponds to the Excel function `=COVARIANCE.P`

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 Reapply
 Advanced

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Get External Data Connections Sort & Filter

A1 : X

	A	B	C	D	E
1	X	Y			
2	34	0.17			
3	29	0.27			
4	29	0.80			
5	30	0.78			
6	30	0.35			

Data Analysis

Analysis Tools

- Anova: Single Factor
- Anova: Two-Factor With Replication
- Anova: Two-Factor Without Replication
- Correlation
- Covariance**
- Descriptive Statistics
- Exponential Smoothing
- F-Test Two-Sample for Variances
- Fourier Analysis
- Histogram